

1. This musical element consists of a piece's sequence of sounds and silences. It controls the length of time between a piece's accents--that is, its major beats. Give the name of this element, whose patterns can be clapped or tapped.

RHYTHM

2. This novel begins in a Flint, Michigan, orphanage in 1936. The title boy leaves his foster home to search for the jazz musician Herman Calloway, who turns out to be his grandfather. Give the title of this Christopher Paul Curtis Newbery winner.

BUD, NOT BUDDY

3. Adding a negative number to another number is the same as completing this basic operation. What operation is the same as adding negative four to positive five?

SUBTRACTION

4. This mixture category includes aerosols, foams, emulsions, and gels. Examples include cream, fog, mayonnaise, and blood. Identify this mixture of two or more particles in which each substance retains its own properties.

COLLOID

5. This Kentucky city is home to the Muhammad Ali Center and Muhammed Ali International Airport. What city was the birthplace of Muhammad Ali and has Kentucky's largest population?

LOUISVILLE (KENTUCKY)

6. A basket of fruit costs \$3.33. The store is selling the baskets for one-third off. How much will the baskets cost on sale?

\$2.22 (TWO DOLLARS AND TWENTY-TWO CENTS)

7. In this picture book, a man takes the Number 5 bus to work each morning. When he wakes up with a cold, his zoo animal friends visit him at home. What is this Caldecott-winning collaboration of Philip C. and Erin E. Stead?

(A) SICK DAY FOR AMOS MCGEE

8. Led by their patriarch known as "Ole Ran'l," this Kentucky family is best remembered for their legendary feud with a family led by its patriarch known as "Devil Anse." Identify this Kentucky family, one half of a longstanding feud with the Hatfields of West Virginia.

MCCOY(S)

9. Composers in this era loved polyphony [puh-LIF-uh-nee], a musical texture with multiple independent voices. The opera, cantata, and oratorio all developed during this period that lasted from about 1600 to 1750. Name this era, whose composers included Handel and J. S. Bach.

BAROQUE (PERIOD or ERA)

10. Free fall is a constant type of this physics vector. Newton's Second Law of Motion states it equals force divided by mass. Name this change in the rate of an object's velocity.

ACCELERATION

11. Be specific. This property is found in nickel, cobalt, and some rare-earth elements. Magnetite - sometimes called lodestone - and iron also have it. Name this strongest type of magnetism.

FERRO(MAGNETISM) (Accept word forms)

12. This instrument family's members can be pitched or unpitched. The maracas, steelpan, timpani, and cymbals all belong to it. Which family consists of instruments that can be struck, scraped, rubbed, or rattled?

PERCUSSION

13. This type of device typically contains a magnetic needle that aids navigation by aligning with the magnetic north pole. What do we call these instruments that typically display cardinal directions and are helpful when you're lost?

COMPASS

14. Two poems of this type are *The Iliad* and *The Odyssey*, both probably written by the ancient Greek Homer. They use heroic similes, typically beginning in the middle of the action, and often include gods and goddesses. Give the word for a long narrative poem about a hero's adventures.

EPIC (POEM)

15. Larry's math class ordered 2 and 1/2 pepperoni pizzas and 3 and 1/2 cheese pizzas. The pizza shop switched the orders, making 3 and 1/2 pepperoni pizzas and 2 and 1/2 cheese pizzas. How much less cheese pizza did they receive than they ordered?

1 (ONE) (PIZZA)

16. This ballet's characters include the Bluebird and the Diamond and Lilac Fairies. At its conclusion, Prince Desiré weds Princess Aurora, whom he awakened with a kiss. Name this Tchaikovsky [chigh-KAWF-skee] ballet.

(THE) SLEEPING BEAUTY

17. In this novel's opening scene, Thomas Small spends his 13th birthday moving to Ohio. His family helps Mr. Pluto and Mayhew trick the Darrows into abandoning the title man's treasure. This is the plot of which Virginia Hamilton novel?

(THE) HOUSE OF DIES DREAR

18. Because this igneous rock fractures with sharp edges, ancient civilizations made cutting and piercing tools from it. It forms quickly from cooled lava that has minimal crystal growth. Identify this shiny black volcanic glass.

OBSIDIAN [ub-SID-ee-un]

19. This term can follow the word "Pennsylvania" in a two-word name for a particular cultural group with German ancestry that settled in the state. What term most often refers to an individual of Germanic ethnic origin native to the Netherlands?

DUTCH

20. There are 23 sugar cookies and 15 chocolate chip cookies in a bowl. What is the ratio of sugar cookies to all the cookies?

23:38 (TWENTY-THREE TO THIRTY-EIGHT) (ACCEPT: TWENTY-THREE OVER THIRTY-EIGHT OR TWENTY-THREE THIRTY-EIGHTHS)

21. When this story element is internal, it pits a character against him- or herself. External types include person versus nature, society, or person. Give the word for the struggle that drives a story's plot.

CONFLICT

22. According to a famous story, this American upholsterer was responsible for convincing George Washington to use a 5-pointed star in the design of the American flag. Though her actual role has been questioned by historians, whose name is now synonymous with the creation of the first U.S. flag?

(ELIZABETH "BESTY") ROSS

23. This type of folk dance is similar to tap dance. It evolved from the hornpipe, reel, and jig in England and Ireland. What is this dance that requires wooden shoes?

CLOG(GING)

24. This atmospheric layer extends from 30 to 250 miles above Earth's surface. Like the mesosphere and thermosphere, it rests between the stratosphere and exosphere. Identify this region of electrically charged particles, which produces auroras and helps spread radio waves.

IONOSPHERE [eye-AH-nus-feer]

25. Jill's desktop measures 1,200 millimeters. How long is her desktop in centimeters?

120 (ONE HUNDRED TWENTY CENTIMETERS)

26. This polygon contains four 90-degree angles. What polygon also has two pairs of opposite sides equal in length?

RECTANGLE

27. In one Aesop fable, this animal lies in a manger and prevents oxen from eating the hay there. In another, a wolf chooses freedom over this animal's life of a collar and chain. In yet another fable, which creature loses his bone while barking at his reflection in the water?

DOG

28. Born Araminta Ross, she was often called "Moses" by those she assisted on the Underground Railroad. Identify the influential American who escaped slavery herself before courageously leading hundreds of other enslaved people to their freedom.

(HARRIET) TUBMAN

29. In the "grand" one of these ballet movements, a dancer does a split in mid-air. The dancer throws her working leg forward while extending her supporting leg backward. Name this jump from one foot to the other.

JETÉ [zhuh-TAY]

30. This astronomer assisted Tycho Brahe and corresponded with Galileo about the telescope. His mathematical discoveries proved Copernicus's heliocentric theory. Who devised the three laws of planetary motion?

(JOHANNES) KEPLER

End of First Half – Begin Three-Minute Inquiry Period

Round 4 - Second Half

Elementary – 2023 District



31. Like plastids, these organelles have their own DNA. They use cellular respiration to produce ATP and small amounts of carbon dioxide. What are these so-called "powerhouses of the cell" that produce energy?

MITOCHONDRIA (Accept MITOCHONDRION)

32. This dancer studied at the American Ballet Theatre School under Merce Cunningham and Martha Graham. Mikhail Baryshnikov performed her choreography in the 1976 *Push Comes to Shove* premiere. Who is this choreographer of *Deuce Coupe* and *Movin' Out*?

(TWYLA) THARP

33. This fairy tale ends with the title characters growing fat from grazing in a meadow. A troll threatens to eat each animal who passes over his bridge, but the largest animal eventually blinds and kills him. What is this fairy tale about a trio of livestock brothers?

(THE) THREE BILLY GOATS GRUFF

34. He was a senator from Illinois before he was elected president by defeating John McCain in 2008. Name the United States' 44th president, for whom Joe Biden served as vice president.

(BARACK) OBAMA

35. A regular hexagon has sides of length 6 units. What is the perimeter in units of the hexagon?

36 (UNITS)

36. In Elizabethan times, this part of the stage got its name because it was tilted toward the ground. Which section of a proscenium stage is found closest to the audience?

DOWN(STAGE)

37. In math, this Greek prefix can be found before "-gon." Identify this prefix meaning "many."

POLY- (Accept stated or spelled)

38. The card shop has 5 different birthday cards, 6 different get-well cards, and 2 all-occasion cards for sale. How many different combinations of cards, one from each category, can be made from the available card selection?

60 (COMBINATIONS)

39. This word describes any vascular plant that reproduces using an ovule - that is, an exposed seed. Many of those seeds are in cones, meaning conifers belong to this group, along with ginkgos and cycads. Identify these opposites of angiosperms [AN-jee-oh-spermz].

GYMNOSPERM(S) [JIM-noh-sperm]

40. This city, which last hosted the Summer Olympic Games in 1984, will next host the games in 2028. Name this American city, the country's 2nd largest by population.

LOS ANGELES (ACCEPT: L.A.)

41. The first half of this compound word is what mammals use to nourish their young. The second half can be a verb meaning "tremble" or "jiggle." What is this compound word for an ice cream drink?

MILKSHAKE

42. In the U.S. government, the holder of this office is second only to the president in authority over the U.S. military. What presidential cabinet member leads the executive department that oversees the U.S. Armed Forces?

SECRETARY OF DEFENSE

43. The Fibonacci sequence is a special number pattern, with each number being the sum of the previous two numbers. What is the fifth term of a Fibonacci sequence that begins 0,1,1?

3 (THREE)

44. This invertebrate class is the largest in the phylum Mollusca [muh-LUS-kuh]. A few species, including conches and limpets, can be eaten. What is this class that also includes snails and slugs?

GASTROPOD(S) [GAS-truh-pod]

45. This artistic tool is usually wooden and comes in A-frame and H-frame varieties. Field ones are small and portable, while studio ones can support works weighing over 200 pounds. Identify this upright stand used to support a canvas.

EASEL

46. A grocer placed 20 bags of apples on the display. He sold X bags of apples. In terms of X, how many bags of apples did he have left in the display?

20 - X

47. During this period, artists branched out from religious work to depict ancient Greek and Roman subjects. Giotto and Donatello were two masters of its early period, while the so-called "High" part lasted from about 1475 to 1525. Name this era of rebirth during which Michelangelo and Leonardo also worked.

(THE) RENAISSANCE

48. Be specific. This framework is found in sponges, starfish, and all vertebrates. It provides more structural support than the type invertebrates have. What is the word for bones that hold an animal from inside the body?

ENDOSKELETON

49. Also known as conscription, this process was last utilized in the United States during the Vietnam War. What term refers to how individuals may be selected to fill military vacancies, sometimes involuntarily?

(THE) DRAFT

50. In a series that already contains commas, this punctuation mark can separate the items in the series. More commonly, it comes between two independent clauses when they lack a coordinating conjunction. What is this punctuation that looks like a period vertically above a comma?

SEMICOLON

51. Juan had \$312.00 in his savings account. He made two deposits of \$5.00 each. How much money does he have in savings now?

\$322 (THREE HUNDRED TWENTY-TWO DOLLARS)

52. This person was the only American-born artist to exhibit in Paris with the Impressionists. *The Letter* and *The Child's Bath* show the influence of Japanese prints. What is the name of this artist known for her portraits of mothers and children?

(MARY) CASSATT

53. In economics, the law of this measurement states that as a product's price increases, fewer people will want to purchase that item. What is this economic measurement often discussed in terms of its relationship with a product's available supply?

DEMAND

54. Scientific name, please. This bone's head forms a ball-and-socket joint with the hip. At the other end, it connects to the knee. Identify this longest bone in the human body, informally called the thigh bone.

FEMUR

55. State the following sentence's appositive phrase: My favorite basketball team, the University of Kentucky, is not doing well this year.

THE UNIVERSITY OF KENTUCKY

56. Give the following sentence's adjective clause: People who are obedient follow the rules.

WHO ARE OBEDIENT

57. This branch of geophysics uses tools called geophones and hydrophones. Some of its scientists try to study volcanic interiors, while others locate gas and oil fields. Identify this science that studies artificial and natural vibrations, including earthquakes.

SEISMOLOGY [size-MAH-luh-jee]

58. Haley was measuring angles with her protractor. One of the angles was 108 degrees. What type of angle is greater than 90 degrees and less than 180 degrees?

OBTUSE (ANGLE)

59. A poem closely associated with this sculpture states, "Give me your tired, your poor, your huddled masses." What sculpture, a gift given from France to the United States, can be seen raising its torch high above the harbor in New York City?

STATUE OF LIBERTY

60. African instruments of this type include the bata, dundun, and djembe [JEM-bay]. Some tribes use the so-called "talking" kind to communicate over long distances. Name this percussion instrument whose membrane head the player strikes with a mallet or the hand.

DRUM(S)

End of Second Half – Begin Three-Minute Inquiry Period

Extra Questions

61. This former Bizaardvark actress more recently portrayed Nini Salazar-Roberts on *High School Musical: The Musical: The Series*. Billboard's 2022 Woman of the Year, she embarked on the Sour Tour after releasing her debut studio album. Who is this singer-songwriter of "Deja Vu" and "Drivers License"?

(OLIVIA) RODRIGO

62. Spelling required. It was ridiculous to excuse her suspicious behavior. Spell the word in the previous sentence that is a synonym of "absurd."

R-I-D-I-C-U-L-O-U-S

63. The academic coach was putting study materials into folders. He put 25 study sheets in the first folder, 31 sheets in the second folder, and 37 sheets in the third folder. How many sheets will he put in the fifth folder if he continues with this pattern?

49 (FORTY-NINE SHEETS)

64. This scientist established the New York Infirmary for Women and Children and, after moving back to England, the National Health Society. She studied at Geneva Medical College, opened a Women's Medical College, and was a professor at the London School of Medicine for Women. Name this first female doctor educated in the U.S.

(ELIZABETH) BLACKWELL

65. Characters introduced in recent years on this long-running children's television program have included Lily, who has experienced homelessness, Julia, who is neurodivergent; and Ji-Young, the program's first Asian American Muppet. Name this television program that featured iconic characters such as Oscar the Grouch, Big Bird, and Elmo for decades.

SESAME STREET